Inherited-F4 pricing jump between `2245b75` and `main`

2026-10-18 · `research/f4_pricing_jump_20261018/` · class: **accounting** (PR #1774) + **algorithm** (PR #1862) · no bug found · no leaderboard row moves

Question

The committed spec of `ECBS1hfd5bf1885227` (`claim-koblitz-rootfind-n13.json`) gave very different inherited-F4 `total_gae` when built from `2245b75` and from `main` on x86-64. Reproduced exactly on this host: `ic-f4-m2` target 0 warm-up 982.628 ⟶ 1,949.912 GAE; `ic-f4-m3` 671.404 ⟶ 25,818.785 GAE.

Method

x86-64 (AMD EPYC 9V74, 4 vCPU), rustc 1.97.0, isolation L0, counts only. `cargo build --release --bin ecbench` at opt-level 1 in one sparse worktree with one shared target directory. Reduced spec `bisect/f4only.json` (arms `rho-strong`, `ic-f4-m3`, `ic-f4-m2`; F4 signature byte-identical to the full spec at both ends). Signature = SHA-256 of the 64 F4 rows' `total_gae`, `lower_bound`, `unpriced`, per-phase `gae`, `solver_ops`, calls, trials, relations, adds, doubles. Recursive bisection over the 303 first-parent commits of `2245b75..origin/main`: 16 points evaluated, two change points, each then split to a single commit.

index	commit	signature	note
0	`2245b7596`	`69a5576f9885`	archived
38 ⟶ 39	`6ea104b4b` ⟶ `cd226d0ce`	`69a5…` ⟶ `6bdf…`	merge of #1774
inner	`3f4a87aae` ⟶ `c8b1594f6`	`69a5…` ⟶ `6bdf…`	single commit
95 ⟶ 96	`e15e29dbd` ⟶ `f2c9202ca`	`6bdf…` ⟶ `becf…`	merge of #1862
inner	`df1fbcf67` ⟶ `d1d034d94`	`6bdf…` ⟶ `becf…`	single commit
303	`abee07ee7`	`becf95c66869`	main

Other evaluated points (37, 41, 46, 56, 75, 94, 98, 103, 113, 151) agree with their neighbours; the full log is `bisect/drive.log`.

Culprit 1: `c8b1594f6` (PR #1774), accounting

`run_pipeline` priced the solver at the pinned `ns_per_word_xor` only for `op_unit == "word XORs"`. Inherited F4 reports `word XORs (elimination, specialisation and linear elimination only)`, so the term was wall-priced, and `ecbench` then removed it as `…_uncharged` (`lower_bound: true`). The commit matches by prefix. Pinned ratio for `K_0 / GF(2^13)` in `docs/ic/calibration.json`: 0.002304 GAE per word XOR, unchanged since before `2245b75`. Across the merge, every F4 row keeps the same `solver_ops`, calls, trials, relations, adds, doubles and non-relations phase GAE. The relations increase divided by `solver_ops` is 0.0023040 on all 64 rows. A correction, not a change of work.

Culprit 2: `d1d034d94` (PR #1862), algorithm

"Harvest root linear rows by default." Where the root gate fires, the multi-variable linear rows join the system, which changes the tree. `ic-f4-m3` target 0 warm-up: reductions 1,886 ⟶ 1,917; propagations 2,800 ⟶ 2,619; eliminated 1,940 ⟶ 1,790; splits 808 = 808; word XORs 10,305,435 ⟶ 10,914,662. Over the 32 m3 rows, after/before word XORs has median 1.0715, range [1.0485, 1.1109]. `ic-f4-m2` is unchanged (gate does not fire). Roots unchanged (all records verified); the counter charges real work. On this workload it is a **regression** of about 7% in m3 relations work, against the PR's "no regression" on its own held-out cells.

state	m2 mean total_gae	m3 mean total_gae	m3 mean word XORs	lower bound
`2245b75`	744.8	712.9	9,399,300	yes
after #1774	1,600.3	22,368.9	9,399,300	no
after #1862 = main	1,600.3	23,937.6	10,080,165	no

Non-warm-up rows, n = 24 per arm; arithmetic means of recorded values.

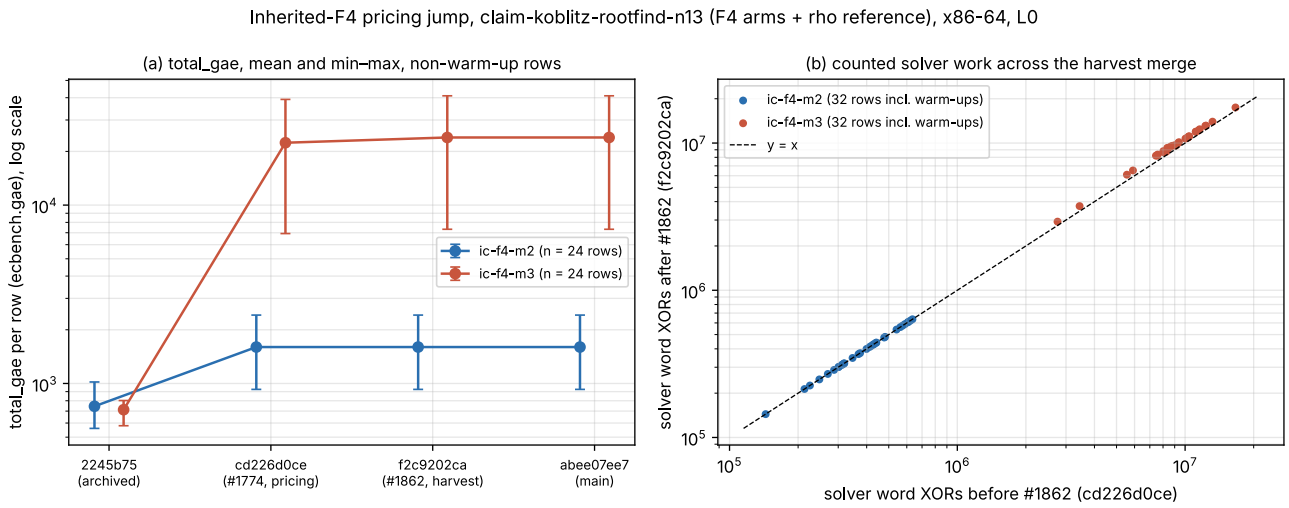


Figure 1: (a) Mean and min–max `total_gae` per arm at the archived commit, after each culprit, and at main. (b) Solver word XORs per F4 row before vs after #1862; m3 sits above y = x. Data `bisect/sigs/`, source `plot.py`.

Affected rows

`docs/ic/leaderboard.json` is built from recorded study files, so no row moves. Tables A–C use no word-XOR-priced arm. Inherited-F4 `S` appears only in the lower-bound diagnostics `f6_ecbench_ladder_diagnostic`, `f4_f5_f6_cross_family_diagnostic` and the frontier replay block, all recorded before #1774. Their specs were re-run with `main` (mean `S`, non-warm-up rows):

session	arm	n	archived S (LB)	main S	main LB?
ladder n13	ic-f4	24	16.27	**488.89**	no
ladder n13	ic-f6	24	861.0	**1,182.3**	no
ladder n19	ic-f4	24	4.916	4.916	yes
ladder n23	ic-f4	24	2.134	**9,479.7**	no
cross K13	inherited-f4	8	16.49	**634.12**	no
cross K13	matrix-f4	8	16.44	**113,830**	no
cross K13	matrix-f5	8	16.44	**73,606**	no
cross R13 / K19 / R19	inherited-f4	8 each	18.45 / 4.957 / 3.368	unchanged	yes

Hazards (accounting): (1) F4-family `S` from before `cd226d0ce` is not comparable with `S` from after it on a curve with a pinned word-XOR ratio; (2) on today's binary the same engine is a complete count on pinned curves (K_0/2^13, K_0/2^23) and a lower bound elsewhere, so a ladder across them mixes the two.

Landed and proposed

Landed: a documentation comment on `is_word_xor_op_unit` (`src/cryptanalysis/ic_framework/mod.rs`); no counted quantity changes. Proposed: version word-XOR-priced IC method identities with the pricing rule; re-measure the #1862 default on the m3 pdp3-Koblitz spec with the harvest as a method parameter (ecbench scrubs `KIC_*` from the child environment).